

Clinical Risk Assessment Report: Patient 16

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 36.1% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.63x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +7.6%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.34 [Bottom 16%] (+7.6%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 2.43 [Top 4%] (+4.8%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.46 [Bottom 10%] (+3.3%)**

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 1.82 [Top 4%] (+3.1%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.23 [Bottom 10%] (+1.9%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: 1.47 [Top 8%] (+1.2%)**

Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

3. Protective / Mitigating Factors (Reducing Risk)

Patient Age (Age) **Value: 67.0 [Avg (64th %ile)] (-2.1%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.28 [Bottom <1%] (-1.5%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Machine Learning Feature Attribution (SHAP Decision Path)

